

Cybersecurity

Backups and Restoring Data with tar

Lesson 5.1



Learning Objectives

By the end of today's class, you will be able to:

- 1 Identify and describe use cases for the three kinds of backups.
- 2 Create (tar) an archive from existing files and directories.
- 3 List and search the contents of an existing archive.
- 4 Extract (untar) the contents of an archive.
- 5 Describe and demonstrate two exploits for the `tar` command.
- 6 Identify and describe how backups are used for compliance.



Welcome

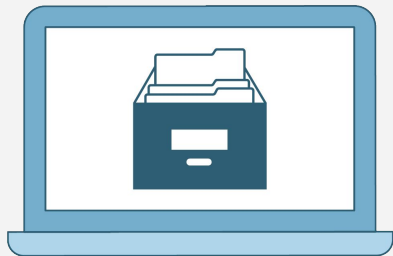


Archiving and Logging Data

Linux tools used by System Administrators

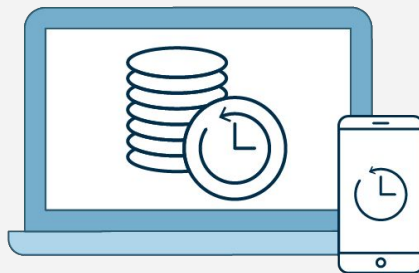
01

Archiving Data



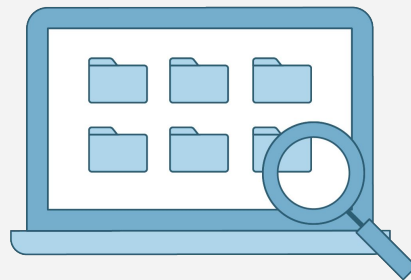
02

Scheduling Backups



03

Monitoring Log Files

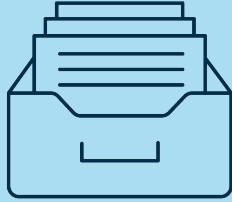


Archiving and Logging Data

These skills are used by system administrators to accomplish the following tasks:

- 1 Overseeing or conducting data backup and recovery.
- 2 Determining how long to retain data and the frequency of backups.
- 3 Monitoring and troubleshooting backups and restoration.
- 4 Providing security for compliance requirements.





Archiving with `tar`

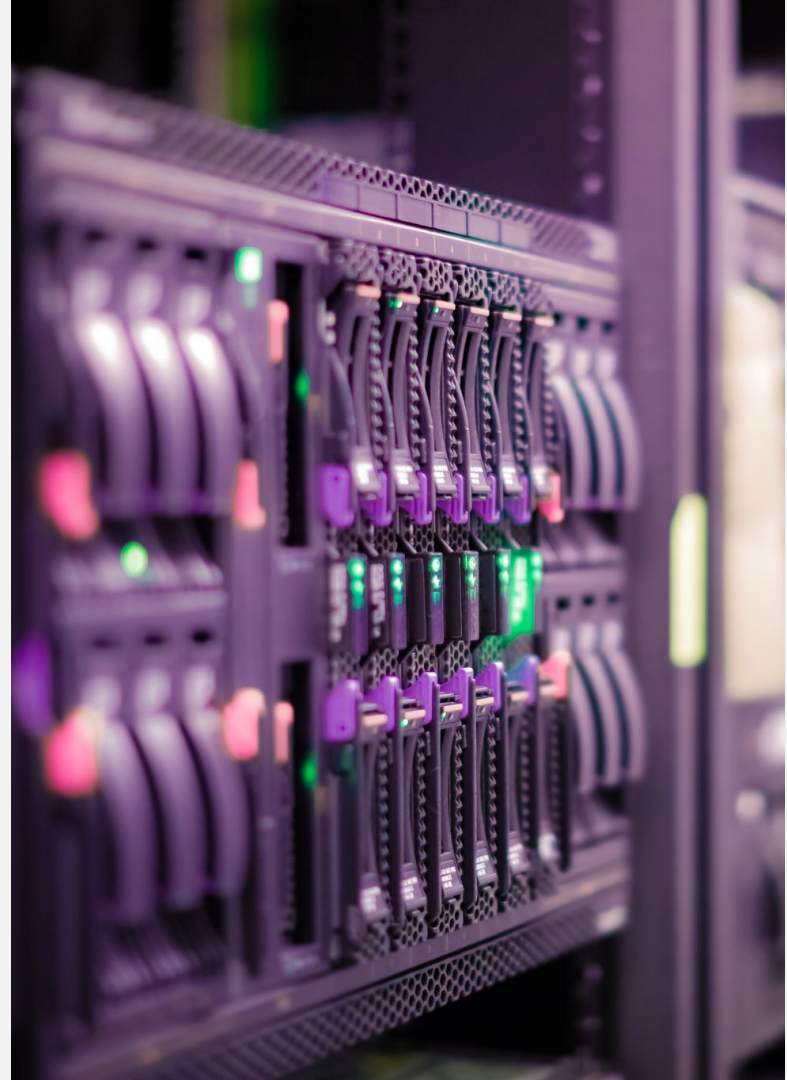
Archives and Backups

Promoting data availability

Backups and data archiving are important to maintain compliance.

Regulations governing IT security industries such as:

- Finance
- Health





Finance

In finance, the common standard for data archiving is the **Sarbanes–Oxley Act (SOX)**. It requires all business records and communication to be retained for five years.

Before the signing ceremony of the Sarbanes–Oxley Act, US President George W. Bush meets with Senator Paul Sarbanes, Secretary of Labor Elaine Chao, and other dignitaries in the Blue Room at the White House on July 30, 2002.



Finance

Markets in Financial Instruments Directive (MiFID II)

Markets in Financial Instruments Directive (MiFID II) requires the European Union's financial firms to retain and reproduce records of all activity from:

- Telephone Conversations
- Electronic Communications
 - Instant Messaging
 - Social Media Interactions





Health

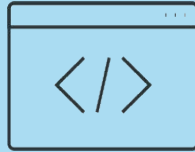
Health Insurance Portability and Accountability Act (HIPAA)

Health Insurance Portability and Accountability Act (HIPAA) requires healthcare providers to keep records for six years.





Most businesses and government agencies have specific **retention** requirements that must be maintained.



Creating and Restoring Backups with `tar`

Cyber professionals must ensure that data is always protected against threats of data loss and interruptions caused by cyberattacks or natural disasters.



Ensuring Availability of Data



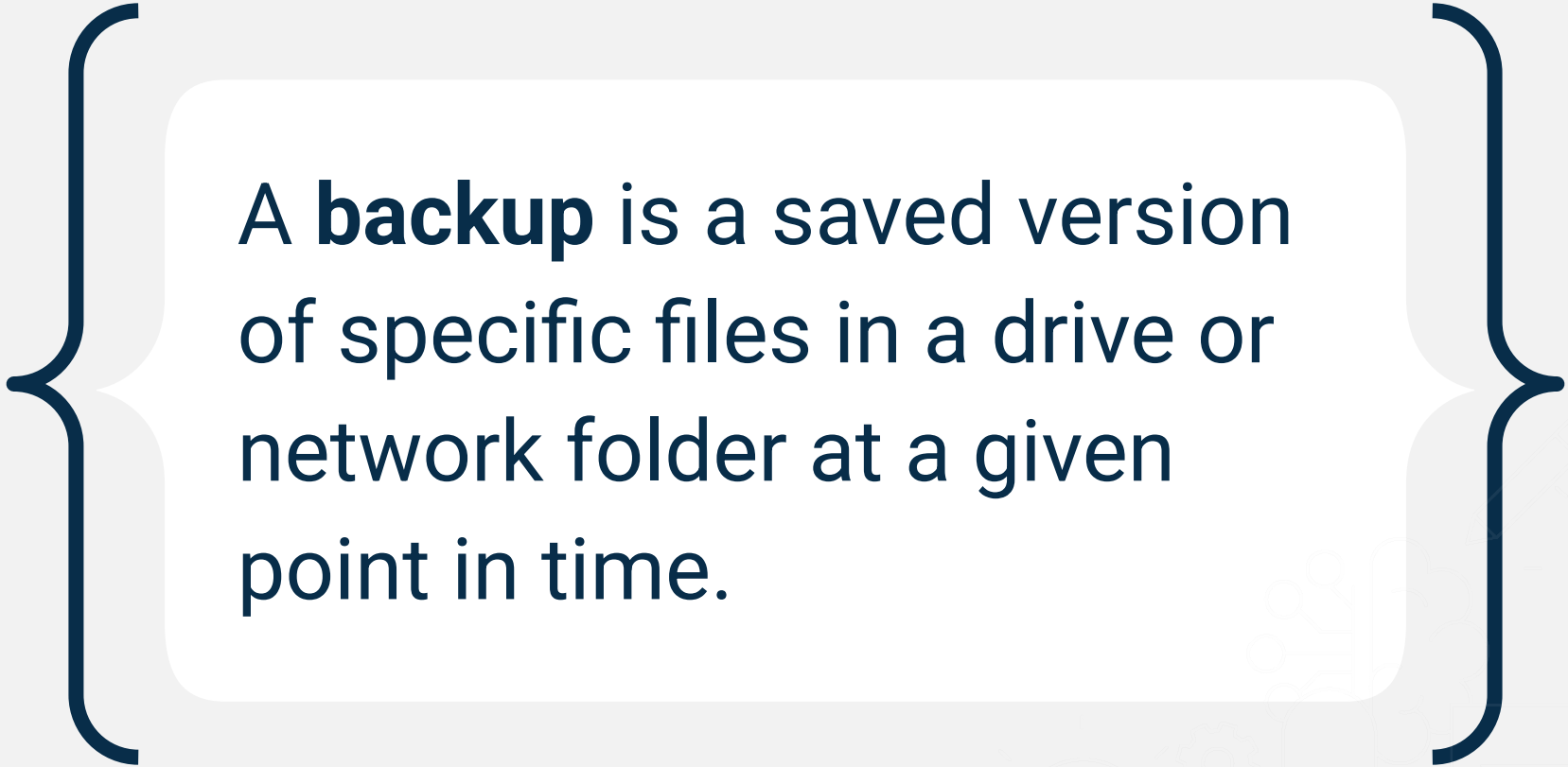
Ransomware

As of July 2023, ransomware payments exceeded \$500 million.



Natural Disasters

In 2019, Hurricane Michael caused massive flooding, destroying hundreds of homes, businesses, and strategic defense data.



A **backup** is a saved version of specific files in a drive or network folder at a given point in time.

Backing Up Data

System administrators backup data so organizations can quickly recover lost assets and restore systems.

- 1 A backup saves specified files on your system and copies them to a safe secondary place.
- 2 Backups are generally completed hourly daily.
- 3 Performing regular backups is a top priority for organizations.



Creating and Restoring Backups

Different Types of Backups

Full Backup

More reliable and offers complete restoration but requires large storage space



Differential Backup

Captures all of the changes made since that initial backup



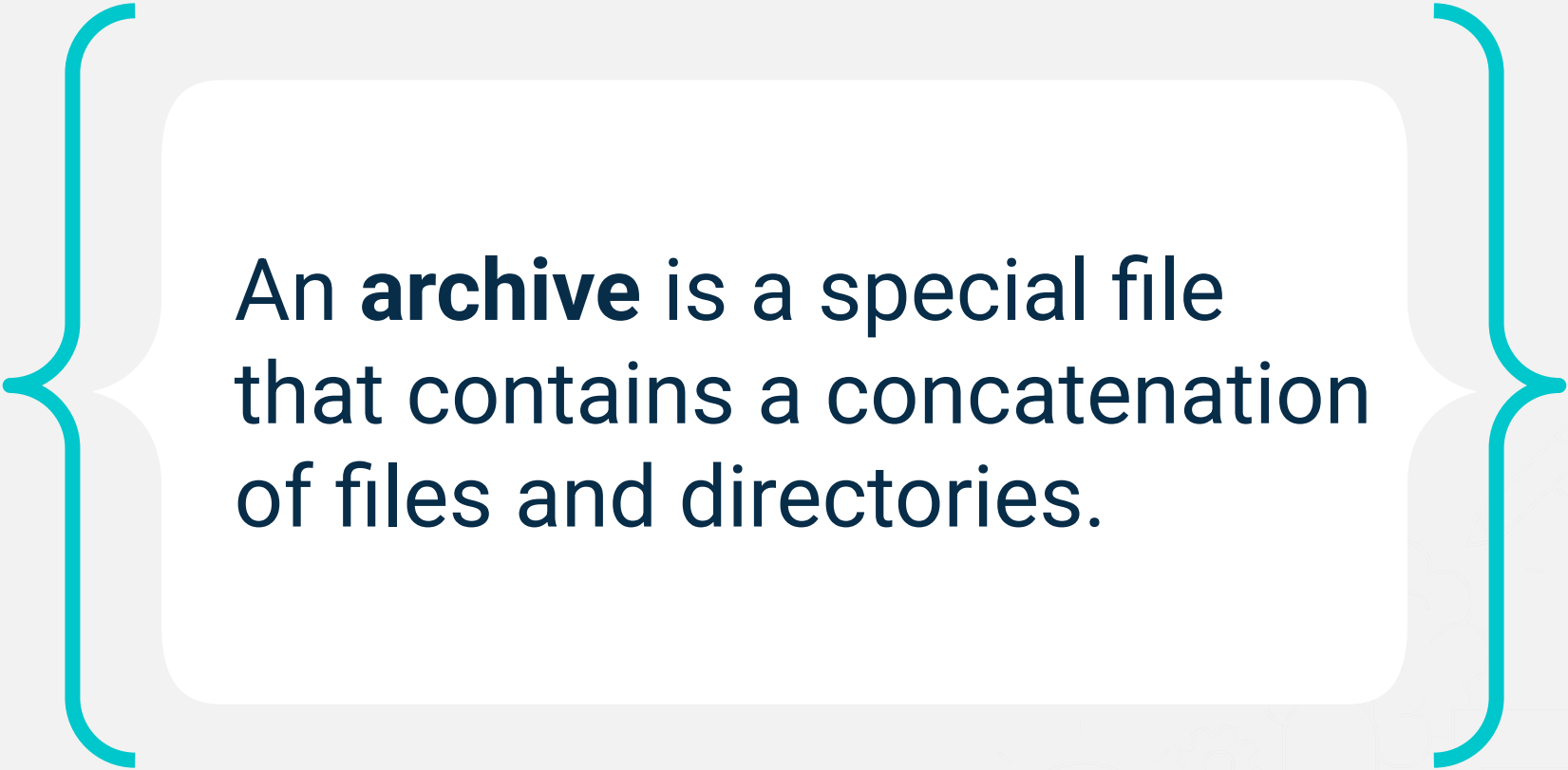
Incremental Backup

Captures only the changes made since the last backup, whether it was a full, differential, or incremental backup





The **tar** (tape archive) command is a Linux utility that system administrators use to create a backup.



An **archive** is a special file that contains a concatenation of files and directories.

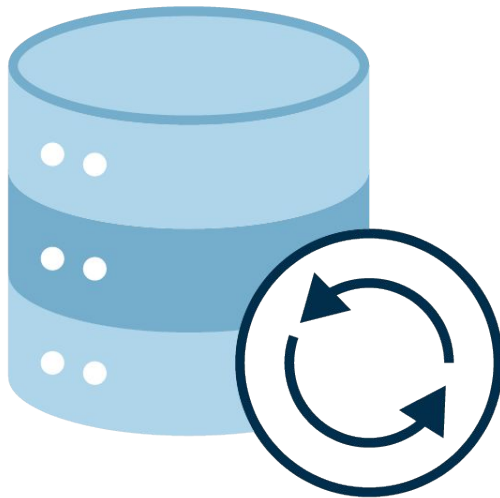


Performing a backup

Ability to “rewind” time

tar takes files we want to backup and creates an archive of them.

Archives created with **tar** are called tarballs, and use the **.tar** extension.



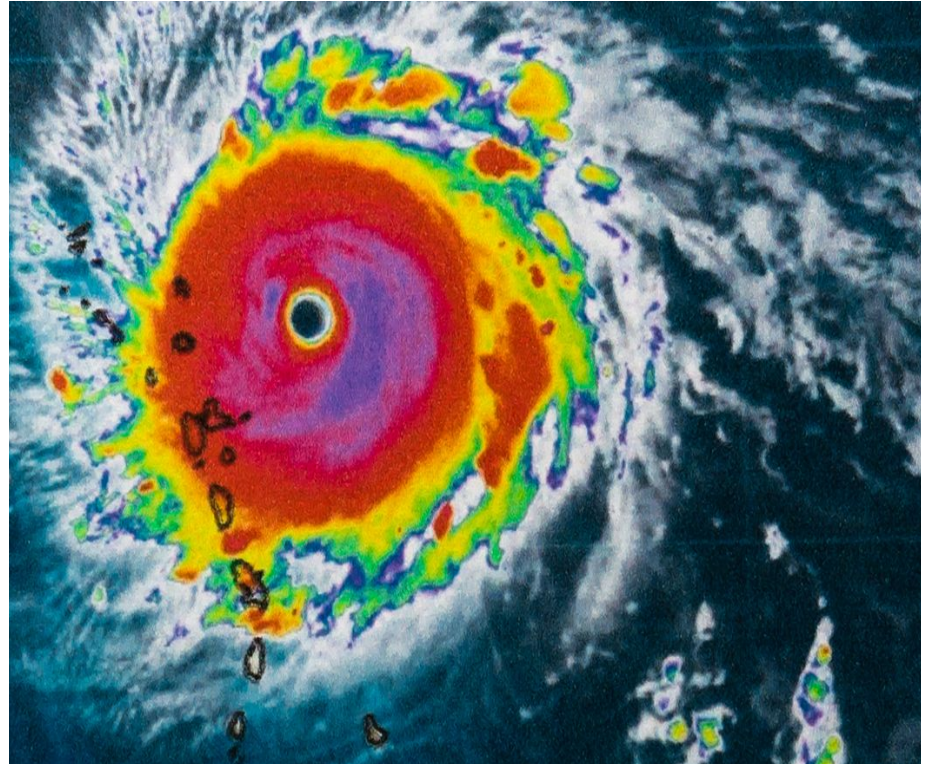
tar backups take up the same amount of space as all the files on the drive.



Scenario

Let's apply this command to the following scenario:

The systems operations center at Tyndall Air Force Base is backing up all system data with the approaching threat of Hurricane Michael.





Creating an Archive

Syntax options for `tar`

Basic syntax of the `tar` command

```
tar [option(s)] [archive_name] [objects_to_archive]
```



Creating an Archive

Create a full backup of all logs in `/var/log`

```
sudo tar [option(s)] [archive_name] [objects_to_archive]
```



`sudo` is required because the files in `/var/log` are system files, so we'll need administrator rights to create a backup.



Creating an Archive

Create a full backup of all logs in `/var/log`

```
sudo tar [option(s)] [archive_name] [objects_to_archive]
```



`tar` is the Linux command to create the backup.



Creating an Archive

Create a full backup of all logs in `/var/log`

```
sudo tar cvf [archive_name] [objects_to_archive]
```



These are short form options:

- **c** stands for create
- **v** stands for verbose
- **f** indicates the title of the archive



Creating an Archive

Create a full backup of all logs in `/var/log`

```
sudo tar cvf 2018-10-12-hurricane-backup.tar [objects_to_archive]
```



`2018-10-12-hurricane-backup.tar` is the name we are assigning to our new archive.



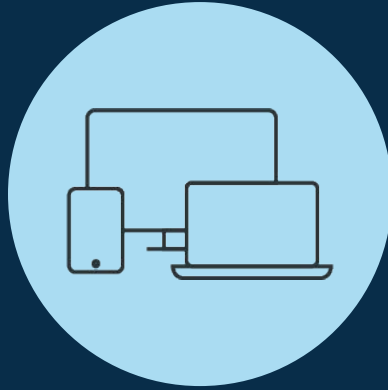
Creating an Archive

Create a full backup of all logs in `/var/log`

```
sudo tar cvf 2018-10-12-hurricane-backup.tar /var/log
```

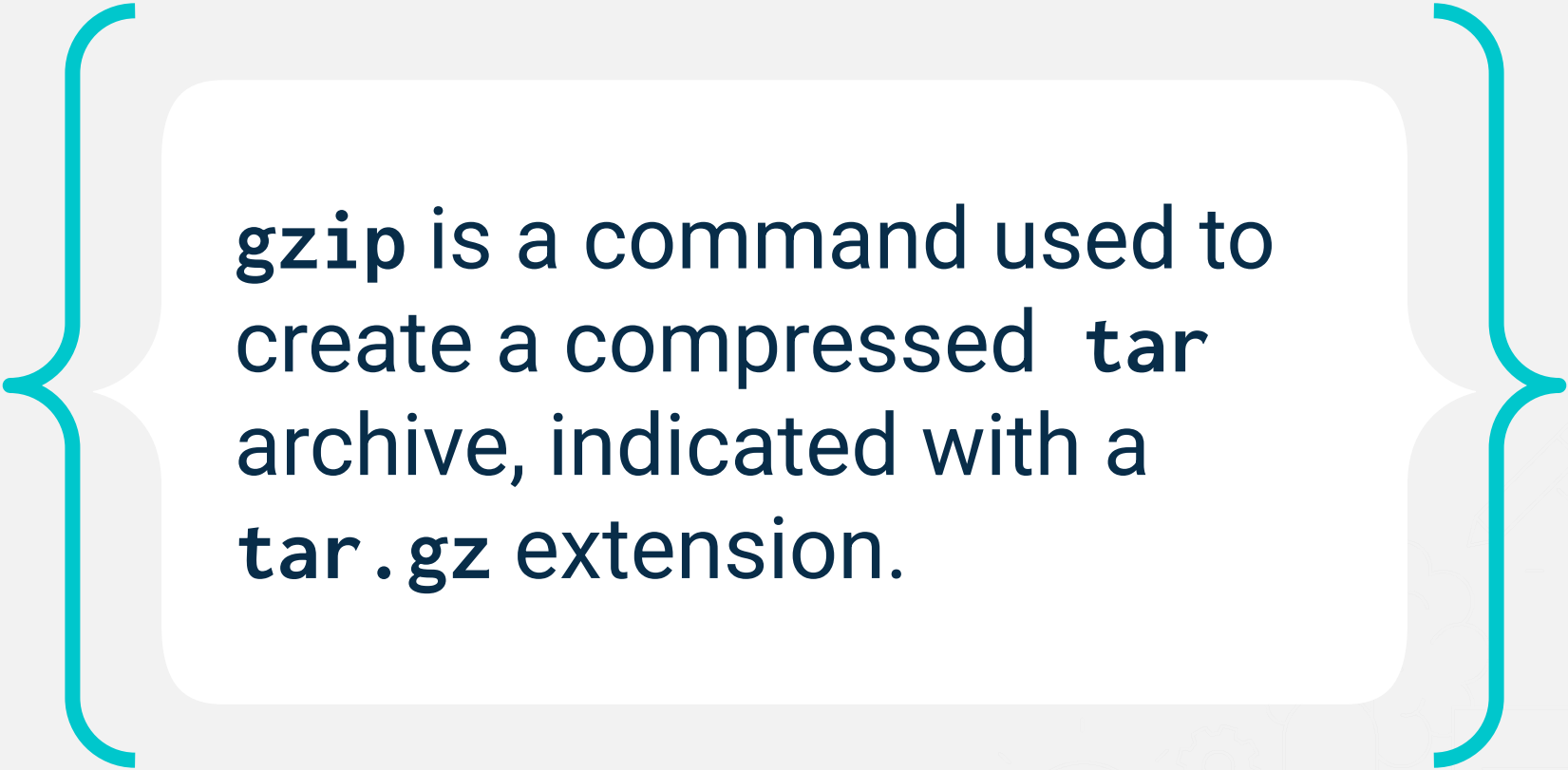


`/var/log` are the objects that we want to archive.



Instructor Demonstration

Running the `tar` command



`gzip` is a command used to create a compressed `tar` archive, indicated with a `tar.gz` extension.

Compressing Archives with gzip



gzip compression is useful when disk space is an issue



Moving files requires less bandwidth, faster transfer times

Compress the archive

```
gzip 2018-10-12-hurricane-backup.tar
```

Extract a compressed archive

```
gunzip 2018-10-12-hurricane-backup.tar.gz
```



Compressing Archives with gzip

Compress & Decompress with **tar**

```
sudo tar czvf 2018-10-12-hurricane-backup.tar.gz /var/log
```



-cz creates the file and compresses it using **gzip**.

```
sudo tar xzvf 2018-10-12-hurricane-backup.tar.gz /var/log
```



-xz extracts the file and decompresses it using **gzip**.



Archiving Best Practices

Maintaining security, privacy, and integrity

- Be sure archives are not writable by an untrusted user.
- Inspect all data, such as passwords, before writing to an archive, since this data will be copied to the archive.
- Pay attention to the diagnostic and exit status of `tar`.



Demonstration Scenario

Creating archives to protect organizations

In 2019, Hackensack Meridian Health systems were compromised by a **ransomware attack**. Attackers encrypted medical data of many patients in the hospital.

The attackers offered to decrypt the data in exchange for payment, effectively ransoming the government's data. Unfortunately, in the case of Hackensack, the hospital had to pay.

N.J.'s Largest Hospital System Pays Up in Ransomware Attack



Author:
Lindsey O'Donnell
December 16, 2019
/ 11:33 am

2:30 minute read

Skip to:

The ransomware attack earlier this month led the hospital system to reschedule surgeries and appointments.

New Jersey's largest hospital system said that it has paid hackers a ransom after a ransomware attack disrupted its services earlier this month.

Hackensack Meridian Health, a \$6 billion non-profit health provider system based in Edison, N.J., operates 17 hospitals, nursing homes and outpatient centers, as well as psychiatric facility Carrier Clinic. The hospital system told media outlets on Friday that it was targeted by a cyberattack on Dec. 2, crippling its computer software systems for nearly five days.



Demonstration Scenario

Creating archives to protect organizations

Affected systems impacted:

- Doctors
- Patients
- Treatments

After taking the infected systems offline, we'll need to:

- Restore the operating system and applications using the full backup.
- Restore email data using the full backup.

Restoring the email data will require using the tar command to:

- List the contents of the latest full backup to locate the email data.
- Extract the email data from the archive to a directory on the new system.

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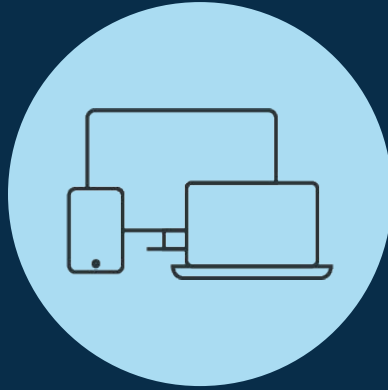
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Instructor Demonstration

Medical Center



Activity:

Creating and Restoring Backups with `tar`

In this activity, you are a junior administrator at Rezifp Pharma Inc. In recent weeks, there have been a series of malware attacks and now the company is strengthening its backup activities.

In response to the malware attacks, your department has decided to create daily full backups of the files associated with the E-Prescription Treatment database that contain (PII).

A pharmacy technician notices that some files in the Patient database are missing and the team discovers that the wrong full backup was used.

Suggested Time:
20 Minutes

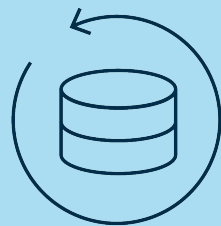


Time's up!
Let's review



Questions?





Restoring Data with **Incremental** Backups

The file sizes for full backups are very large, because they include everything on a system.

Depending on the size of the file system, doing a full backup takes a very long time.



Incremental Backup

Incremental backups are completed after a full backup is performed on a system, only capturing what has changed since the last incremental backup.

1 Incremental backups store a list of which files have changed in a snapshot file, with the extension **.snar**.

2 The snapshot is created when the admin creates the initial level-0 backup.

3 Every time an incremental backup completes, a new snapshot is created that contains only files that have changed since the last full or incremental backup.



Incremental Backups

Incremental backups drawbacks:

Restoration

Restoring the backup is slow, as you must restore the latest full backup plus all the previous incremental backups.

Loss of Data

Loss of data possible if a single backup is corrupted.

Additional Verification

Verify an archive was created with no error using `tar -W`.

Creating the Incremental Backup

Incremental backups are a form of **tar** archive

- **tar** stores additional metadata and information
- Tracks modified and deleted files since the last backup
- Incremental backup will only contain modified files
- Stored in a snapshot file **.snar**





Creating an Incremental Backup

Syntax options for incremental backups with `tar`

```
tar cvvWf emerg_back_sun.tar --listed-incremental=emerg_backup.snar --level=0 emergency
```



`--listed-incremental=emerg_backup.snar` tells `tar` that the backup will be part of a series of incremental backups, and stores metadata about the backup in a snapshot file `emerg_backup.snar`.



Creating an Incremental Backup

Syntax options for incremental backups with `tar`

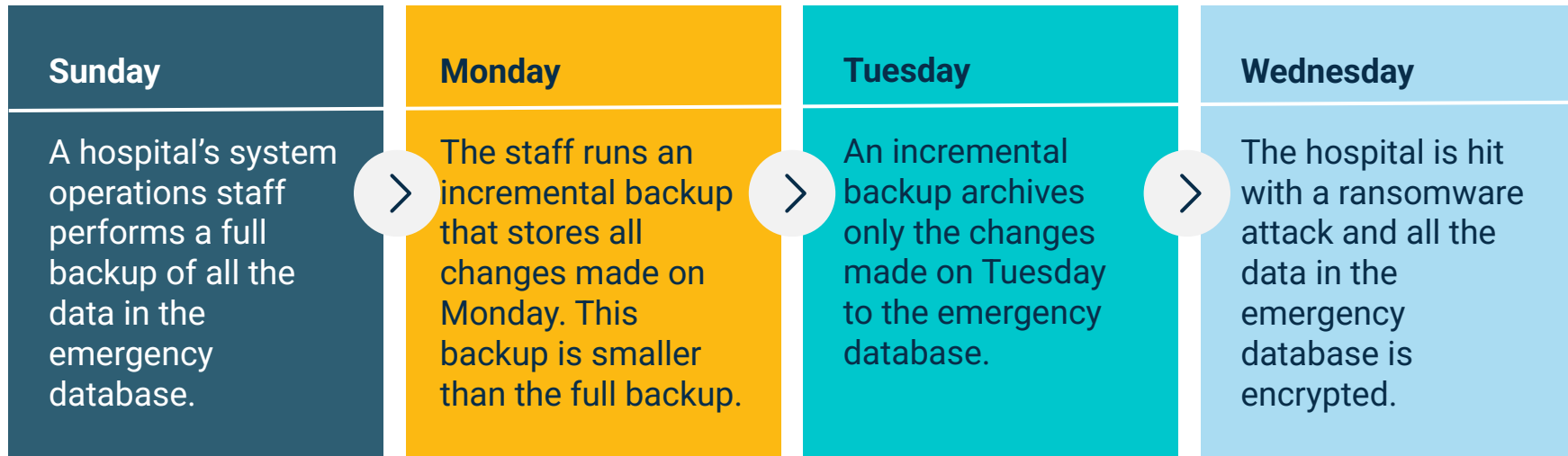
```
tar cvvWf emerg_back_sun.tar --listed-incremental=emerg_backup.snar --level=0 emergency
```

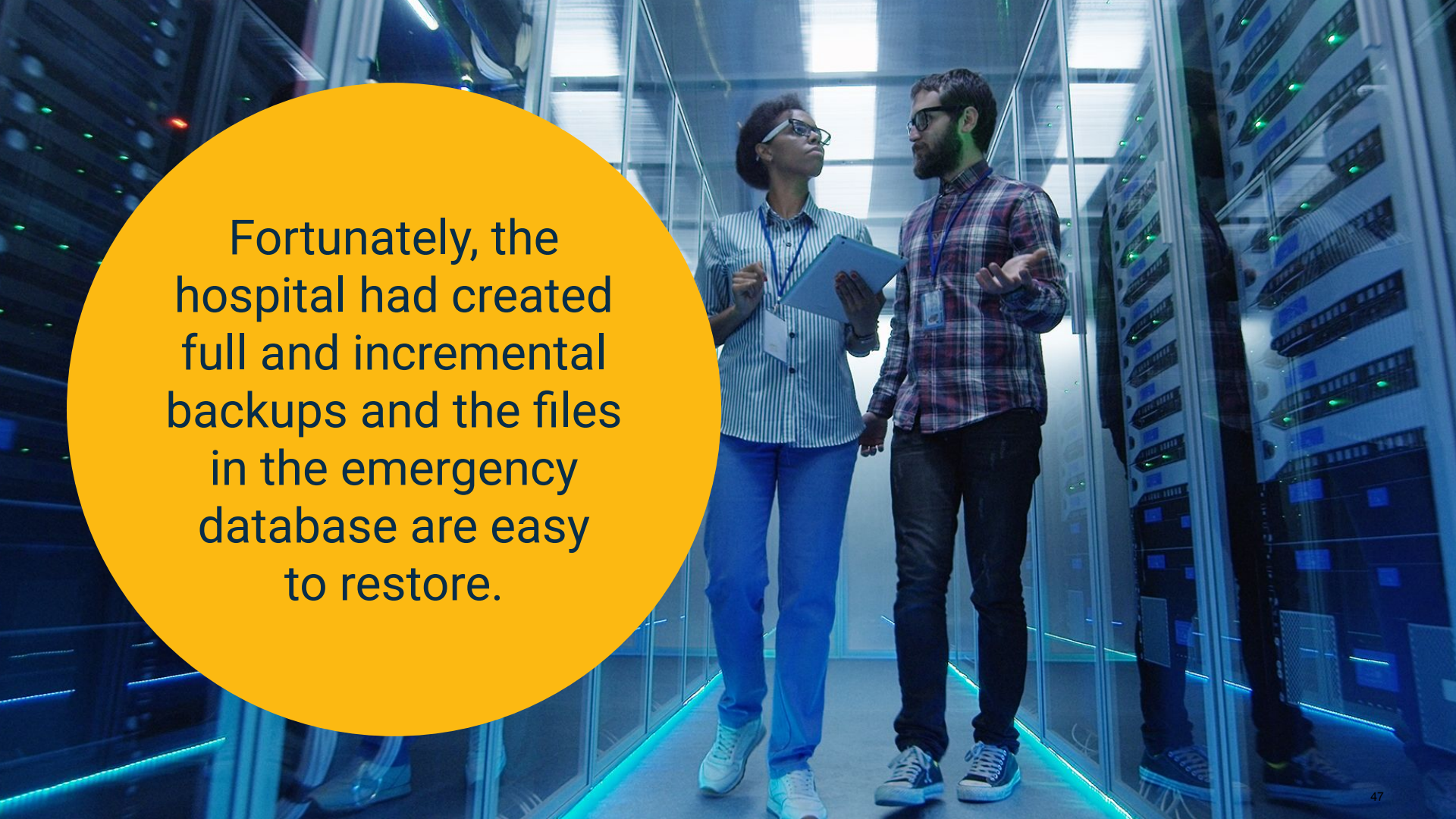


`--level=0` tells `tar` that this is the first backup of the series.



Incremental Backup Scenario



A man and a woman, both wearing glasses and lanyards, are walking through a server room. The woman is holding a tablet and pointing at it. The man is gesturing with his hands. They are surrounded by rows of server racks with glowing green lights. A large yellow circle is overlaid on the left side of the image, containing text.

Fortunately, the hospital had created full and incremental backups and the files in the emergency database are easy to restore.



Incremental Backup Scenario

01

Create a full backup of the emergency directory and a snapshot file.

02

Display the contents of the level 0 backup generated on Sunday.

03

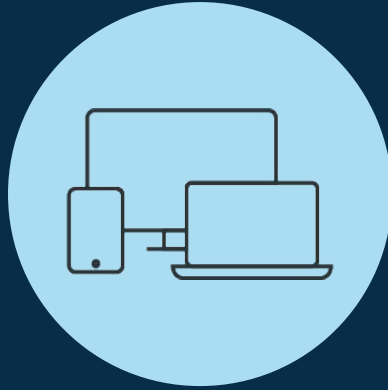
Create the incremental backups generated on Monday and Tuesday.

04

Remove the emergency directory to mimic the ransomware attack on Wednesday.

05

Restore the emergency directory.



Instructor Demonstration

Incremental Backups



Activity:

Restoring Data with Incremental Backups

In this activity, you will continue as a junior administrator at Rezifp Pharma Inc.

Your organization has just adopted a new backup policy and you are responsible for validating the new process.

In this simulation, you will perform two tests:

- Restore patient files that were deleted during an attack on Sunday.
- Create two new files and perform an incremental backup on Monday to ensure the new files were properly backed up.

Suggested Time:
20 Minutes



Time's up!
Let's review

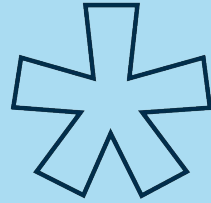


Questions?



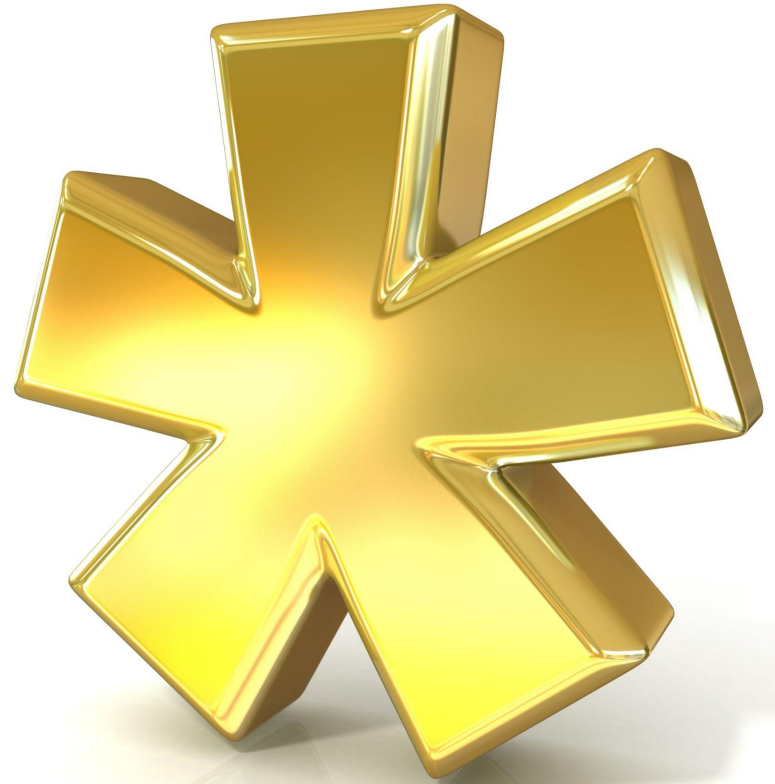


Break
15 mins



Exploiting the `tar` Command with the Checkpoint and Wildcard Options

Hackers can combine the `tar` command with the wildcard character (*) and the checkpoint feature to plant malware on a system.





Using Wildcard with tar

Wildcards and Archives

```
~$ ls Documents/ExploitTar
```

```
f1 f2 f3 f4 f5 f6 f7 f8 f9
```

```
~$ tar cvf archive.tar ./*
```

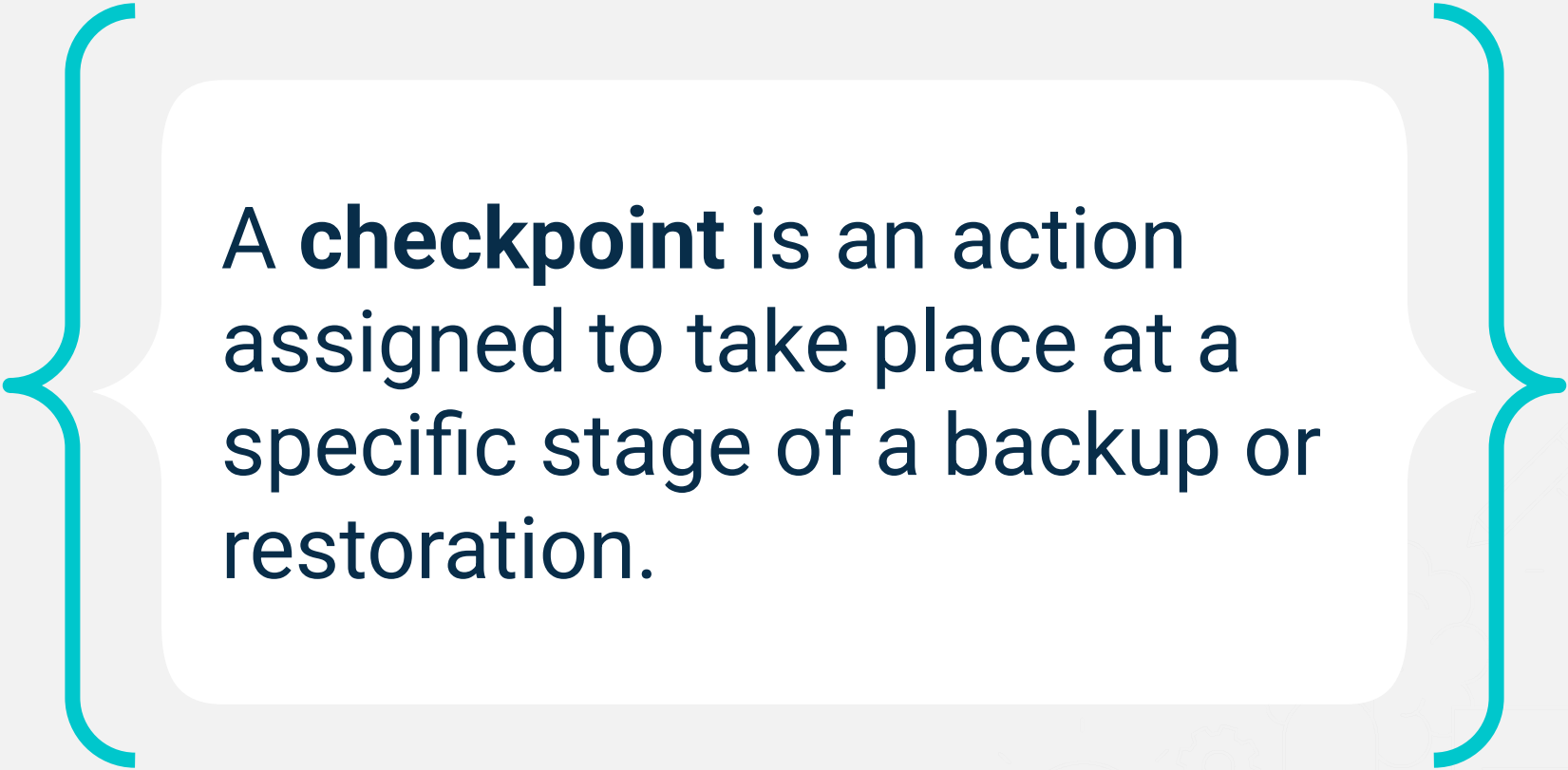
We previously used wildcard characters
* for creating archives.

* is used to indicate all files in a
directory

For example:

Documents/ExploitTar contains multiple
directories that we want to archive.

./* represents all the contents of
Documents/ExploitTar



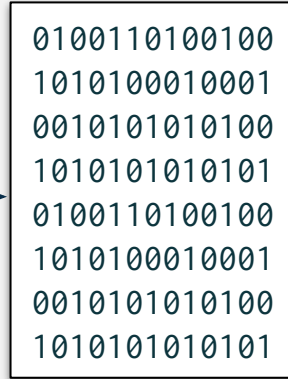
A **checkpoint** is an action assigned to take place at a specific stage of a backup or restoration.

Using Checkpoints

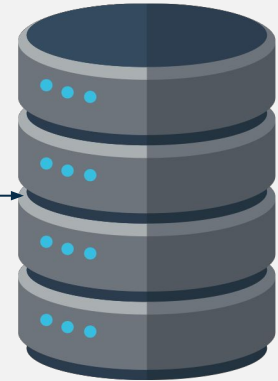
For example:



Once the backup reaches 1,000 files, a checkpoint triggers the backup to pause and display the amount of remaining disk space.



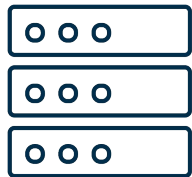
When restoring files from a remote site, a message will print the number of bytes transferred every 5,000 files.



The action taken and the frequency will vary based on the circumstance.

Using Checkpoints

Including checkpoints helps to avoid serious issues



Space Availability

Help avoid the accidental usage of all the freespace on a backup server.



Network Performance

Reduce the amount of data transferred over network at one time.

Using Checkpoints

Checkpoints involve two commands. The first defines the checkpoint, and the second specifies the action to be taken at that checkpoint:

`--checkpoint=1000`

- Indicates how often the checkpoint occurs.
- At every thousandth file, an action will take place.
- Syntax: `--checkpoint=[n]`

`--checkpoint-action=du`

- Indicates the action that will take place.
- `du` displays the available disk space on the machine.
- Syntax: `checkpoint=[ACTION]`

Exploiting System Vulnerabilities

Previously we discussed how sysadmins harden against exploits

Identify Weaknesses

- Identify weak or vulnerable services
- Exploit with open-source code or use their own
- ACE



Arbitrary Code Execution

- ACE is extremely dangerous
- Attempt to gain root privileges



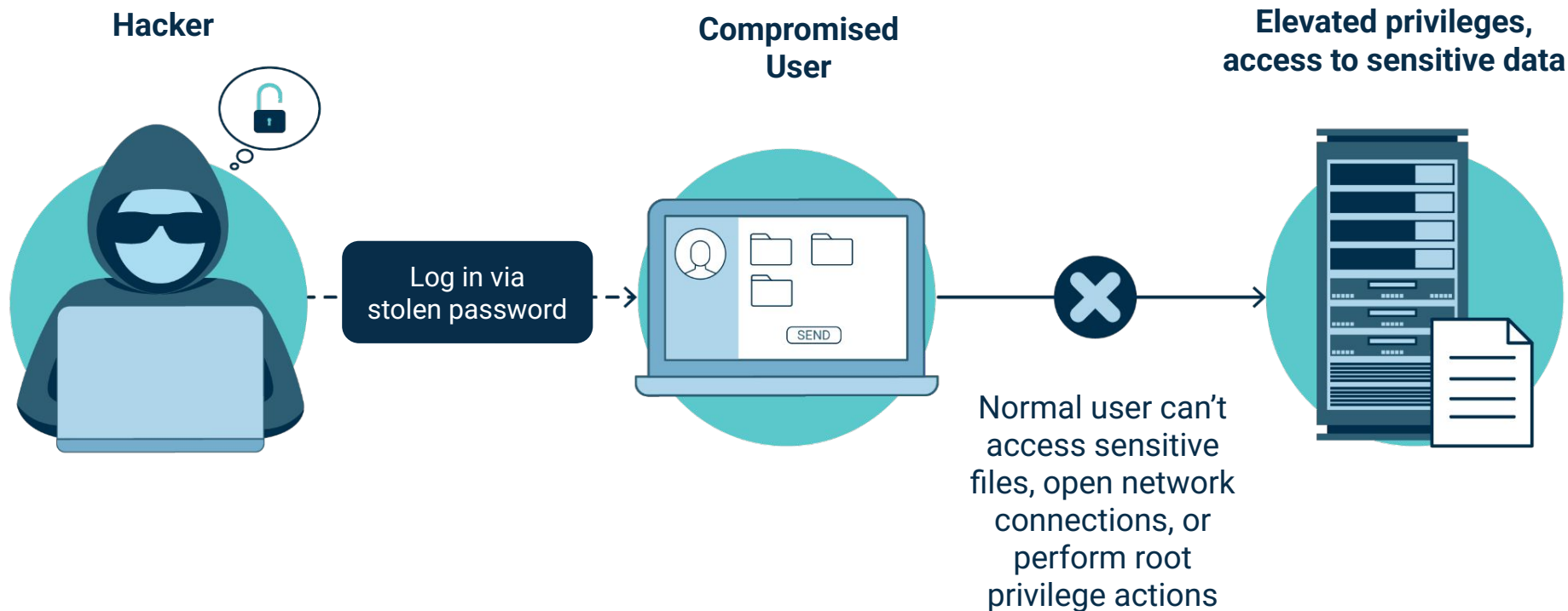
tar checkpoints

- Known vulnerability
- Plant malicious code on a system
- Allows root privileges



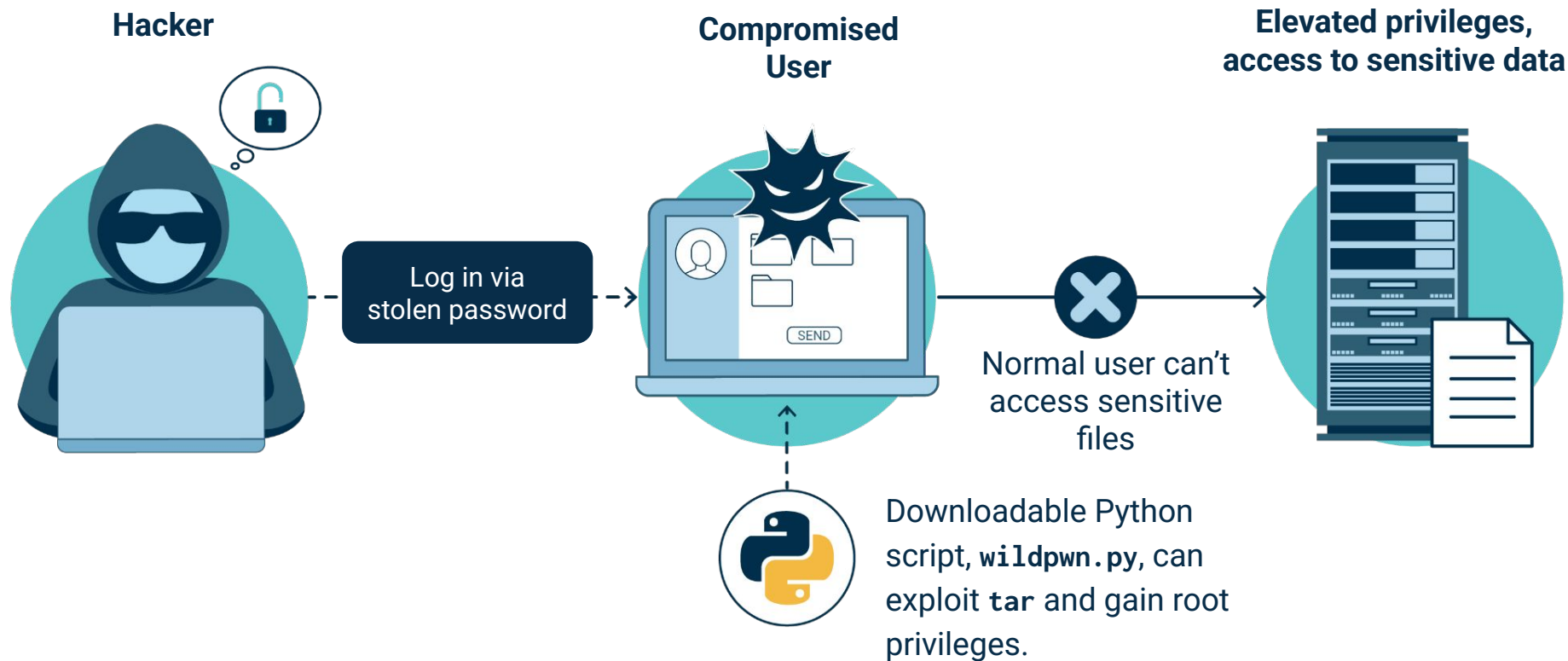
Checkpoint + Wildcard Exploit

First, a hacker gains access to a system by impersonating a normal user:



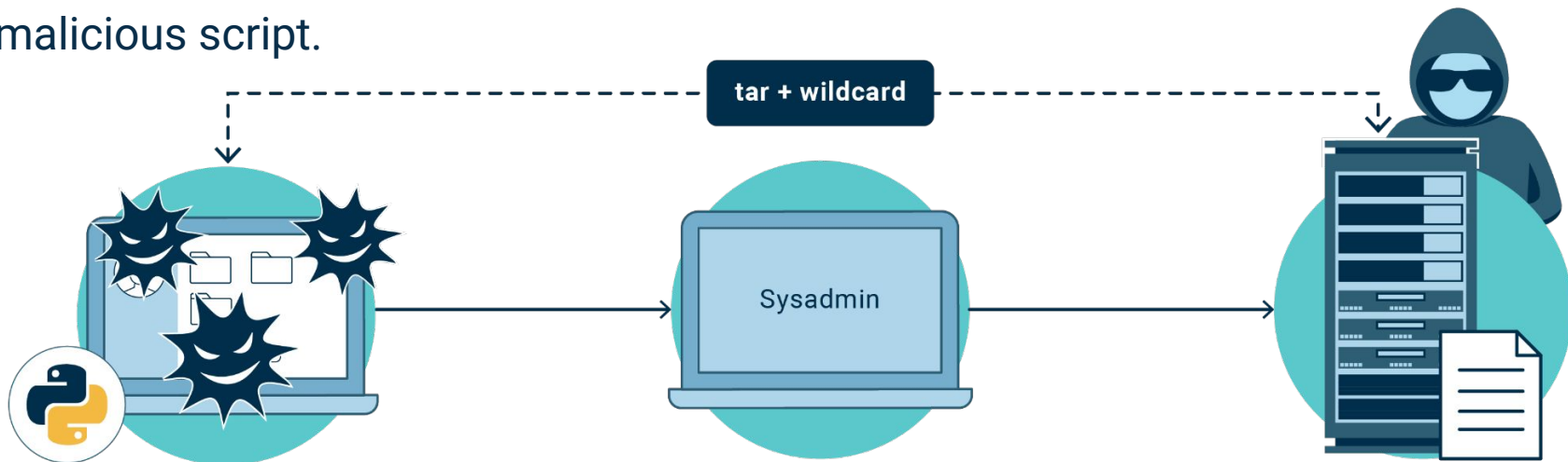
Checkpoint + Wildcard Exploit

A normal user can typically download and execute files from the internet. If a file is malicious, it can compromise the system.



Checkpoint + Wildcard Exploit

Unsuspecting admins with root privileges run a `tar` command containing a wildcard. This will create an archive containing the directory of the malicious script.



`wildpwn.py` plants three malicious files on the machine: one hidden, and two files with names like `tar checkpoint` commands.

Sysadmin unknowingly creates an archive containing planted malicious script.

A hacker can drop into root shell, gaining full access to the system.



But how do **checkpoints**
and **wildcards** fit into
this exploit?



Checkpoint + Wildcard Exploit

Exploiting tar

- When an attacker runs `wildpwn.py`, it creates three files.
- The attacker can create these files with **normal** user privileges.
- Files include a **malicious** script (`.webscript`) and two files resembling checkpoint commands:
 - `--checkpoint=1`
 - `--checkpoint-action=exec=sh .webscript`

After adding the first file to archive, **tar** will execute `.webscript`.





Demonstration Scenario

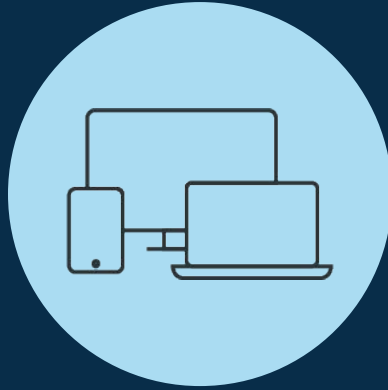
Exploiting tar

An attacker has gained access to our system by stealing the password for the instructor user.

Although, for this example, we will pretend that instructor user does not have root access. In your activity, you will truly be working on a user without root access.

We will be taking the role of the attacker and will use `wildpwn.py` to plant malicious files. These files will be executed when an administrator with root privileges runs the `tar` command.





Instructor Demonstration

Exploiting tar



Activity:

Exploiting the `tar` Command with the Checkpoint and Wildcard Options

In this exercise, you'll play the part of a penetration tester hunting for vulnerabilities in a target system's backup procedures.

- You'll use the `wildpwn.py` tool to verify that the system's current configuration is indeed vulnerable to this kind of privilege escalation attack.
- Then, you'll research and suggest two mitigation strategies to protect the server in the future.

Suggested Time:
25 Minutes



Time's up!
Let's review



Questions?





Recap

Backups and Restoring Data

01

Identified and described three types of backups

02

Created, listed, and searched archives

03

Extracted contents of an archive

04

Used exploits with the **tar** command



Next class

cron and Scheduled Jobs



Thank you